

Integrals – 7

Computing an integral without an antiderivative

11 сентября 2026 года

Theoretical minimum.

Fact 1 (the checklist). None of the integrals below has an elementary antiderivative, so the answer must come from the integral as a whole. Five questions, in this order: does the interval carry a symmetry $x \mapsto a + b - x$, $x \mapsto 1/x$ or $x \mapsto a/x$ (Integrals – 2)? does the integrand satisfy a functional equation, so that adding a few copies of the integral kills the unknown (Theorem 1)? can a parameter be inserted so that the derivative in it is elementary (Integrals – 3)? is the integrand a power series in disguise (Integrals – 4)? is it a beta or gamma integral (Integrals – 5)? If none applies, the integral is probably not to be computed at all but estimated, and one looks at where the mass sits (Fact 3).

Theorem 1 (substitutions of finite order). Let φ map the interval of integration onto itself with $\varphi^m = \text{id}$ and $\int f \circ \varphi = \int f$ for every f . Then summing the m copies of the integral obtained from $x, \varphi(x), \dots, \varphi^{m-1}(x)$ replaces the integrand by its φ -average, which is often constant. The involutions $x \mapsto a + b - x$ on $[a, b]$ and $x \mapsto 1/x$ on $(0, \infty)$ (with the invariant measure $\frac{dx}{x}$) are the case $m = 2$; the Möbius map $x \mapsto 1 - \frac{1}{x}$ has order 3 and permutes the intervals $(-\infty, 0)$, $(0, 1)$, $(1, \infty)$.

Theorem 2 (Cauchy–Schlömilch). For $a, b > 0$ and any f for which one side converges,

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \frac{1}{a} \int_0^\infty f(x^2) dx.$$

The proof is the substitution $x \mapsto \frac{b}{ax}$ and the half-sum of the two forms. In particular $x^2 + x^{-2} = \left(x - \frac{1}{x}\right)^2 + 2$ turns a symmetric exponent into a Gaussian.

Fact 2 (recognising a series). A power series under an integral sign is worth summing first: $\sum_{k \geq 0} \frac{(-1)^k x^{2k+1}}{2 \cdot 4 \cdots (2k)} = xe^{-x^2/2}$ and $\sum_{k \geq 0} \frac{x^{2k}}{(2 \cdot 4 \cdots (2k))^2} = I_0(x)$. Conversely, an integral over a long list of terms is worth reading as one coefficient of a generating function.

Fact 3 (where the mass sits). For a large power the integral $\int_0^c x^r g(x) dx$ is governed by a neighbourhood of the right endpoint: if g is continuous and $g(c) \neq 0$ then $\int_0^c x^r g(x) dx \sim \frac{c^{r+1} g(c)}{r}$, and if $g(c) = 0$ the next term of the Taylor expansion takes over, shifting the answer by one power of r . This is the crude form of Laplace's method; the precise version is in Limits – 6.

1. (Putnam 1953) From the identity

$$\int_0^{\pi/2} \ln \sin 2x \, dx = \int_0^{\pi/2} \ln \sin x \, dx + \int_0^{\pi/2} \ln \cos x \, dx + \int_0^{\pi/2} \ln 2 \, dx,$$

deduce the value of $\int_0^{\pi/2} \ln \sin x \, dx$.

2. (Putnam 1980) Evaluate $\int_0^{\pi/2} \frac{dx}{1 + (\tan x)\sqrt{2}}$.
3. (Putnam 1982) Evaluate $\int_0^\infty \frac{\arctan(\pi x) - \arctan x}{x} dx$.
4. (Putnam 1985) Evaluate $\int_0^\infty \frac{e^{-1985(t+t^{-1})}}{\sqrt{t}} dt$. You may assume that $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.
5. (Putnam 1964) Find all continuous positive functions $f(x)$ for $0 \leq x \leq 1$ such that

$$\int_0^1 f(x) dx = 1, \quad \int_0^1 x f(x) dx = \alpha, \quad \int_0^1 x^2 f(x) dx = \alpha^2,$$

where α is a given real number.